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21 April 1981
SUPERSEDING
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MILITARY SPECIFICATION

INFLATION SYSTEM EQUIPMENT, COMPRESSED AIR, FOR MARK 5, MOD 2 (15 PERSON) AND MARK 6 (25-PERSON) INFLATABLE LIFEBOATS

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers compressed air inflation equipment for inflatable lifeboats.

1.2 Classification. Equipment shall be of the following types, as specified (see 6.2.1).

Type I - Cylinder capacity 250 cubic inches (in^3).
Type II - Cylinder capacity 455 in^3.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-A-1034 - Air, Compressed, For Breathing Purposes.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Depart ment of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116 - Preservation-Packaging, Methods of.
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof Flexible.
MIL-C-17415 - Cloth, Coated, and Webbing, Inflatable Boat and Miscellaneous Use.
MIL-I-22110 - Inhibitors, Corrosion, Volatile, Crystalline.
MIL-I-45208 - Inspection System Requirements.
MIL-W-83420 - Wire Rope, Flexible, for Aircraft Control.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-889 - Dissimilar Metals.
MS22054 - Valve, Life Raft, Flush Type, Topping Off (For Multiplace Life Rafts).
MS26545 - Cylinders, Compressed Gas, Non-Shatterable.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
803-4382177 - 25-Person Inflatable Lifeboat MK 6.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE AGENT
Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC. AGENT
National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA TRAFFIC Dept., 1616 "P" Street, N.W., Washington, DC 20036.)

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DEPARTMENT OF TRANSPORTATION (DOT)

Code of Federal Regulations, Title 49, CFR
170-199 - Hazardous Material Regulations.

(Application for copies should be addressed to Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. The inflation equipment furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.2 Sample for first article inspection. Prior to beginning production, samples shall be examined and tested as specified in 4.5 (see 6.4).

3.3 Materials and components. Materials and components shall conform to applicable specifications as listed or required herein.

3.3.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3.2 Metals. All metal parts shall be corrosion-resistant metals. Dissimilar metals, as defined in MIL-STD-889, shall not be used in intimate contact with each other.

3.4 Design and construction.

3.4.1 Cylinders. Compressed air cylinders, 5000 pounds per square inch (lb/in^2) working pressure, shall be the type as specified by the contracting activity (see 6.2.1).

3.4.1.1 Cylinder volumes. Cylinder volumes shall be as follows:

(a) 15-person lifeboat MK5 - 250 in^3.
(b) 25-person lifeboat MK6 - 455 in^3.

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3.4.2 Cylinder valve assembly. The cylinder valve assembly shall operate freely under all conditions of atmosphere and humidity through a temperature range of minus 20°F to 165°F. It shall have no parts easily removable by hand and shall meet the following requirements:

(a) Rated working pressure. 5000 lb/in^2 at 70°F.
(b) Leakage rate. A maximum of 3 cubic centimeters (cm^3) of free air per hour (maximum) at 70°F ± 2°F at sea level atmospheric pressure (14.7 lb/in^2).
(c) Discharge rate. A minimum of 99 percent of 455 in^3 cylinder volume charged to 5000 lb/in^2 in 45 seconds (or less) at all temperatures in the range from minus 20°F to 165°F.
(d) Size, overall. Shall not exceed a cylindrical volume 2-3/4 inches diameter, 5-inches long.
(e) Valve and cylinder interface. Threaded (see 3.4.2.4(b)).

3.4.2.1 Valve design. Specific requirements for valve design shall be as follows:

(a) Remote operation. By pull cable from outside of the life-boat container.
(1) Pull, generally parallel to the longitudinal centerline of the cylinder.
(2) Maximum pull force 20 pounds.
(3) Maximum travel 2 inches, to fully open the valve.
(4) Cable shall pull free after fully opening the valve.
(5) Cable shall be capable of assembly into the valve after the cylinder has been fully charged and all other valve connections made.
(6) Indicator(s) shall be provided to signify that the valve is in the "open" and "closed" positions. Indicator may be "dots" or "pointer(s)", however, in either case, protected from damage. Where indicating "dots" are used, the color "red" shall signify "open" and "green" "closed".
(7) Pull cable outlet port shall be internally threaded 5/8-18 UNF-2B to receive the threaded end of the conduit (see 3.4.2.7).

3.4.2.2 Body material. Body material shall be corrosion-resisting steel (CRES) 300 series.

3.4.2.3 Valve outlet. Valve outlet requirements shall be as follows:

(a) Externally threaded 3/8-18 NPSM, total full thread length 9/32-inch, minimum.
(b) Outlet end shall be faced perpendicular to the centerline of the external thread.
(c) Minimum outlet gas passage orifice, 7/32-inch diameter.
(d) An integral anti-recoil configuration (see 3.4.2.5).

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3.4.2.4 Cylinder interface port. The cylinder end of the body of the valve shall be configured as follows:

(a) At least two parallel "wrench flats", 1-1/2 inches ± 0.01 inch apart having no less than 7/8-inch land.
(b) Internally threaded 1-1/8-12 UNF-3B, 0.844 inch plus 0.000 inch, minus 0.044 inch deep to receive the cylinder neck thread, detail "X" (see MS26545).
(c) The tapped hole shall terminate with a faced land, 1-inch diameter, 90 degrees true to the perpendicular centerline of an internal thread. The center area of the plane may have an 11/16-inch diameter recess configured to receive the valve poppet seat or to allow for access to the poppet mechanism. The purpose of this land is to accommodate the soft copper sealing washer (see 3.4.2.11).
(d) A 1/8-inch diameter vent hole shall be drilled through one side of the tapped hole, centered approximately 3/16 inch below the sealing washer flat, perpendicular to the vertical centerline of the valve.

3.4.2.5 Safety fittings. Safety fittings shall be as follows:

(a) Anti-recoil. The valve outlet (see 3.4.2.3) shall be configured to eliminate recoil of the cylinder. Design of fitting shall be as shown on figure 1.
(b) Frangible disc and components. A safety port, closed with a threaded plug and replaceable safety disc and washer shall be provided. The threaded plug shall be provided with discharge outlets so arranged that in event of rupture of the frangible disc, compressed air will discharge to atmosphere in a pattern to preclude recoil of the cylinder. Seating of the disc in the valve shall be with the disc between the plug and the washer.
    (1) Disc shall meet the following:
        a. Material - Brass or copper, gold plated both sides to preclude corrosion.
        b. Diameter 9/16 inch.
        c. Burst pressure - Disc shall fail at 6500 lb/in^2 ± 300 lb/in^2 when seated in the valve port, with its phenolic washer and plug (thickness not specified intentionally).
    (2) The washer shall meet the following:
        a. Outside diameter - 19/32 inch.
        b. Inside diameter - 9/32 inch.
        c. Thickness - No less than 1/32 inch.
    (3) The threaded plug shall be configured for installation, utilizing conventional wrenches (either standard hex or recessed socket hex). Assembly into the valve shall require torque defined as "tight" or "snug".
(c) Safety wire. Provision shall be made for attachment of a safety wire with lead seal to the valve body (see 3.4.8).

3.4.2.6 Charging. The valve assembly shall be so configured that when assembled with the cylinder, the cylinder can be fully charged through the valve outlet. Special recharging fittings and tools, if required, shall be furnished in quantities specified (see 6.2.1).

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3.4.2.7 Protective conduit. A flexible "conduit", one end threaded 5/8-18 UNF-2A, similar to that in Drawing 803-4382177 shall be provided, screwed into the cable outlet port of the valve. The purpose of the conduit is to prevent fouling of the actuating cable in its passage through the packed boat. The conduit shall be 13-1/2-inches long, for the 15-person boat, type I equipment, and 30-inches long for the 25-person boat, type II equipment.

3.4.2.8 Weight. The completed valve assembly, not including the flexible conduit (see 3.4.2.7), shall not weigh in excess of 2 pounds.

3.4.2.9 Valve actuating mechanism pull cable interface. The valve actuating mechanism shall be configured to accept the actuating cable specified in 3.4.7 and as shown on Drawing 803-4382177. In addition, all the components of the mechanism shall allow assembly in only one mating position or orientation. The part of the assembly (cable drum) interfacing with the pull cable shall be clearly marked to identify the proper location of the pull cable hook when the valve is in the closed position and ready for firing. The cable drum shall also be so designed that an incorrect reverse or "backward wrap" of the pull cable on the cable drum will still allow the mechanism to function properly, and the valve to be fired in a conventional manner.

3.4.2.10 Actuating mechanism cover. A removable cover shall be provided to allow for threading the actuating cable (see 3.4.7) onto the internal operating mechanism. It shall be configured so as to allow assembly in only one position or orientation. Removal of the cover shall not compromise the integrity of the valve assembled on a fully charged cylinder. The cover shall be included in the indicator scheme (see 3.4.2.1(a)(6)).

3.4.2.11 Cylinder valve gasket. Each valve assembly shall be supplied with a soft annealed copper gasket 15/16-inch outside diameter (o.d.), 5/8-inch inside diameter (i.d.), and approximately (no less than) 1/16-inch thick.

3.4.2.12 Labeling. Labeling may be accomplished by indentation (insofar as no damage to the valve parts occurs) or corrosion-resisting steel plate attached by rivets, screws, or drive screws. Care shall be taken to prevent loosening and "back-out" of the fastenings. Information to be provided shall be as follows:

(a) Valve model and serial number (provided by the manufacturer).
(b) Valve working pressure (5000 lb/in^2 at 70°F).
(c) Manufacturer's name and address.
(d) Date of manufacture.
(e) National stock number (to be provided by the contracting officer).

3.4.3 Flexible hose. The connection between the cylinder valves and the inlet manifolds shall be by means of a flexible hose and the necessary end fittings. Hose shall be Aeroquip part number FA6693EGG0241, or equal. The hose and associated fittings shall withstand the low temperature, both inside and outside, created by the passage of high velocity air produced when activating charged cylinders (see 3.6.1).

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3.4.3.1 90 degree fittings. There shall be a 90 degree fitting, Aeroquip part number FF1438-0606B, or equal, placed between the cylinder valve outlet and the flexible hose. There shall be a 90 degree fitting, Aeroquip part number 2089-4-4B, or equal, placed between the manifold and the flexible hose.

3.4.4 Manifold. The manifold shall perform the following functions and include the following fittings:

(a) Be so arranged to assure simultaneous inflation of the two compartments to which they are attached. The difference in the pressure between adjacent compartments shall be 0.3 lb/in^2 or less.
(b) Have a threaded opening for attachment of the fittings on the delivery end of the flexible hose.
(c) Be fitted with nonreturn valves to prevent the passage of the inflation medium from one compartment to the other or to the open air.
(d) Shall not leak when a back pressure of 5.0 lb/in^2 ± 0.1 lb/in^2 is applied to the two inlet diffusers.

3.4.4.1 The manifold shall project not more than 1-1/2 inches from the outer surfaces of the lifeboat. The manifold shall be set in a molded neoprene patch of such composition that it can be readily vulcanized to the type 16A cloth, of MIL-C-17415, of the lifeboat flotation tubes. All exposed metal parts of the manifold shall be covered with molded neoprene to prevent damage to the lifeboat cloth. The manifold shall permit ready attachment and removal of flexible hose connecting it to the valve assembly.

3.4.4.2 Inlet diffusers. The manifold shall contain two diffusers, one at each end of the manifold, and each shall be an integral part of the manifold body. Each diffuser shall have not less than three outlet holes which shall be drilled at an angle to the manifold base. Discharge holes shall be so located that recoil in any direction shall be kept at a minimum, when the air is discharged into the boat inflation tube.

3.4.4.3 Replacement. Replacement of any expendable parts shall be possible from the ends of the manifold body. Removable end plugs shall be provided to facilitate the removal of expendable parts.

3.4.5 Topping-off valve. Topping-off valve shall be in accordance with MS22054.

3.4.6 Relief valves. Relief valves shall open at or below 3.8 lb/in^2 and close at or above 2.3 lb/in^2. The valve body shall be as small as practical and shall be mounted on a flat molded neoprene patch that can be easily vulcanized to the lifeboat flotation tubes. All metal parts which can cause damage to the lifeboat fabric shall be covered with molded neoprene.

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3.4.7 Actuating cables. Valve actuating cables shall be 1/16-inch diameter, 7 by 19 corrosion-resistant steel conforming to MIL-W-83420 aircraft cord. The associated fittings shall be of the swaged type configured for ready attachment to and removal from the cylinder valve actuating mechanism and the lanyard operating device in the lifeboat container. Each actuating cable shall be in accordance with Drawing 803-4382177 and shall withstand a pull of 100 pounds. The lengths required shall be determined from the applicable lifeboat specification and drawings.

3.4.8 Safety wiring. The actuating cables or valve actuating mechanism shall be secured to the cylinder valve body with a wire seal, breaking strength 10 pounds, to prevent accidental inflation. Safety wire shall be installed on the lifeboats prior to shipment of the lifeboats. A red tag shall be attached to the excess safety wire and shall contain the following information:

"WARNING

WEIGH CYLINDER IMMEDIATELY BEFORE INSTALLING ON LIFEBOAT. DO NOT INSTALL IMPROPERLY CHARGED CYLINDER OR IMPROPERLY SAFETIED OR UNSAFETIED INFLATION GEAR ASSEMBLY."

3.5 Interchangeability. All parts having the same function shall be directly and completely interchangeable with each other with respect to installation and performance. The cylinder valve shall be interchangeable with any cylinder regardless of the manufacturer. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength.

3.6 Tare weight. Weight of each empty cylinder valve assembly (without actuating cable or protective conduit) shall be determined to an accuracy of plus or minus 0.01 pound, and the following information in 1/2-inch high letters shall be stencilled on the side of each cylinder.

"Tare weight----------Pounds
Weight of compressed air---------Pounds
Total----------Pounds."

3.6.1 Cylinder charging.

3.6.1.1 Cylinder corrosion protection. Immediately prior to assembly of cylinder valve to cylinder, each cylinder shall be treated with a corrosion inhibitor which meets the active constituent requirements of MIL-I-22110, type II, except that the material shall be in tablet form. CORTEC Corporation tablet 560T, or equal, is satisfactory. The tablet shall be sized in accordance with the manufacturer's instructions for the appropriate volume and corresponding surface area.

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3.6.1.2 Charging of cylinder valve assembly. Each cylinder valve assembly shall be charged with breathable compressed air conforming to BB-A-1034 and having a dew point at minus 20°F or less. The assemblies shall be charged, so that at 70°F ± 2°F, the following nominal pressure and weight are obtained:

Size of cylinder	Compressed air weight	Nominal gage pressure
(in^3)	(lb)	(lb/in^2)
250	2.96 ±0.04	5000
455	5.70 ±0.04	5000

3.7 Identification of product. All items shall be legibly and permanently marked in accordance with MIL-STD-130.

3.8 Workmanship. All parts, components, and assemblies of the inflation system including valves, hoses, fittings, castings, forgings, molded parts, stampings, seals, and machinings shall be clean and free of sand, dirt, pits, scale, rough surfaces, nicks, burrs, sharp edges, cracks, embedment of foreign matter or other harmful, extraneous materials. Edges and covers not otherwise limited by tolerances and finishes necessary for satisfactory function shall be chamfered or rounded. Final assemblies consisting of basic components shall not have evidence of misalinements, cross threading, or poor and improper seatings which would cause damage to incased or enclosed parts, or leakage to occur.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facility suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection system. The contractor shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with MIL-I-45208.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.4).
(b) First article inspection (see 4.5).
(c) Quality conformance inspection (see 4.6).

4.4 Qualification inspection. Qualification inspection shall be conducted at a laboratory satisfactory to NAVSEA and shall consist of the examination and tests specified in 4.4.1.

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4.4.1 Samples for qualification examination and tests. Three cylinder valve assemblies and three each of the following components shall be tested as specified in 4.4.2 through 4.4.4.

(a) Disc and washer.
(b) Hose assembly with fittings.

All components shall be examined as specified in 4.7.1 and 4.7.2.

4.4.2 Cylinder valve assembly. Cylinder valve assembly shall be tested as follows:

(a) Hydrostatic (see 4.9.5).
(b) Leakage rate (see 4.9.6).
(c) Operating force (see 4.9.7).
(d) Operational (see 4.9.8).
(e) Coupling nut torque (see 4.9.9).
(f) Endurance (see 4.9.10).

4.4.3 Disc and washer. Disc and washer shall be subjected to the burst test (see 4.9.1) to determine conformance with 3.4.2.5(b).

4.4.4 Hose assembly with fittings. Hose assembly with fittings shall be proof tested as specified in 4.9.4 to determine conformance with the requirements of 3.4.3 and 3.4.3.1.

4.5 First article inspection. First article inspection shall consist of the examination and tests specified in 4.5.1.

4.5.1 Samples for first article inspection. The manufacturer shall submit three valve assemblies and three each of the following components for the tests specified in 4.5.2 through 4.5.7. All components shall be examined as specified in 4.7.1 and 4.7.2.

(a) Disc and washer.
(b) Hose assembly with fittings.
(c) Manifold assembly.
(d) Topping-off valves and relief valve assembly.
(e) Cable.

The samples shall be representative of the construction, workmanship, components, and materials to be used during production. The first article samples shall be forwarded to the Officer-in-Charge, Annapolis Laboratory, David W. Taylor Naval Ship Research and Development Center, Annapolis, MD 21402. The samples shall be plainly identified by securely attached durable tags marked with the following information:

"Samples submitted by (name) (date) for first article inspection in accordance with the requirements of Contract No._____."

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4.5.1.1 First article test reports. The contractor shall prepare a first article test report in accordance with the data ordering document included in the contract or order (see 6.2.2). Information supplied by the testing laboratory shall be used in the preparation of the report.

4.5.2 Cylinder valve assembly. Cylinder valve assembly shall be tested as specified in 4.4.2.

4.5.3 Disc and washer. Disc and washer shall be subjected to the test specified in 4.4.3.

4.5.4 Hose assembly with fittings. Hose assembly with fittings shall be tested as specified in 4.4.4.

4.5.5 Manifold assembly. Manifold assembly shall be subjected to the patch adhesion test (see 4.9.11) to determine conformance with 3.4.4 and a leakage test to determine conformance with 3.4.4 (see 4.9.3.3.).

4.5.6 Topping-off valve and relief valve assembly. Topping off valve and relief valve assembly shall be subjected to the leakage test (see 4.9.3) and to the patch adhesion test (see 4.9.11) to determine conformance to the requirements of 3.4.5 and 3.4.6.

4.5.7 Cable. Cable shall be tested as specified in 4.9.2 for pull to determine conformance with 3.4.7.

4.6 Quality conformance inspection.

4.6.1 Sampling for quality conformance inspection.

4.6.1.1 Lot.

4.6.1.1.1 Components. For purposes of sampling, a lot shall consist of all components of one kind made under the same conditions and from the same materials.

4.6.1.1.2 Inspection for preparation for delivery. For purposes of sampling, a lot shall consist of one fully prepared shipping container of individual components of one kind fully prepared for delivery, except for sealing, from essentially the same materials and components.

4.6.1.2 Sampling for visual examination. Each valve assembly and each component in the lot shall be visually examined as specified in 4.7.1.

4.6.1.3 Sampling for dimensional examination. A random sample of valve assemblies and components shall be selected from each lot in accordance with level S-2 of MIL-STD-105. The acceptable quality level is 4 percent defective. Examination shall be as specified in 4.7.2.

4.6.1.4 Sampling for burst test (frangible discs and washers). A random sample of burst discs shall be selected in accordance with the sampling procedure of MIL-STD-105, level S-2.

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4.6.1.5 Sampling for all other components. Each of the other components shall be tested as follows:

Component	Test
Cable	Pull (see 4.9.2).
Topping off valve and relief valve assembly	Leakage (see 4.9.3).
Hose assembly with fittings	Proof (see 4.9.4).
Cylinder valve assembly	Hydrostatic (see 4.9.5).
	Leakage rate (see 4.9.6).
	Operating force (see 4.9.7).
Manifold, topping off valve and relief valve assembly	Patch adhesion and leakage (see 4.9.11).

4.7 Examination.

4.7.1 Visual examination. Each valve assembly and component (see 4.6.1.2) shall be examined visually for defects, for conformance to manufacturer's drawings and the requirements of this specification. Classification of defects shall be in accordance with table I.

TABLE I. Classification of defects.

Categories	Defects
Critical	
1	Material not as specified.
2	Type not as specified.
3	Component threads (inlet, outlet) not as specified.
4	Safety port not as specified.
5	Wrench flats not as specified.
6	Safety device not as specified.
7	Dimensions exceed allowable sizes specified.
8	Marking not as specified.
9	Parts or components loose, missing, detached or incomplete.
10	Workmanship not as specified.
11	Safety wiring not accomplished (when applicable).
12	Safety wire not as specified.
Major	None defined.
Minor	
201	Any identification marking missing, incorrect, illegible, or incomplete.
202	Red warning tag missing, incorrect, illegible or incomplete.

4.7.2 Dimensional examination. Each sample valve assembly and component selected in accordance with 4.6.1.3 shall be dimensionally examined to determine conformance to the manufacturer's drawings and to the requirements of this specification. If any measurement deviates from the dimensional requirements specified, the valve assembly or component shall be rejected.

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4.8 Test conditions.

4.8.1 Atmospheric conditions. Unless otherwise specified herein, tests shall be conducted at an atmospheric pressure of 28 to 32 inches of mercury, at a temperature of 77°F ± 18°F and a relative humidity of 80 percent or less. When the tests are conducted at an atmospheric pressure or temperature different from the above values, proper corrections shall be made for the change in instrument reading.

4.8.2 Pressure test. Unless otherwise specified (see 6.2.1), all pressure tests are to be performed using oil-free and moisture-free air of a quality equal to the requirements of BB-A-1034.

4.9 Tests.

4.9.1 Burst. Frangible discs and washers shall be tested to determine conformance to the requirements of 3.4.2.5(b). The discs may be mounted in any convenient manner duplicating the conditions within the cylinder valve. Test pressure may be applied hydrostatically.

4.9.2 Pull. Prior to final assembly into the charged cylinder valve assembly, each actuating cable shall be tested to determine conformance with 3.4.7. Any cable in the lot which fails to pass the test shall be rejected.

4.9.3 Leakage.

4.9.3.1 Topping-off valve. Topping-off valve shall show no leakage when a back pressure of 5 lb/in^2 is applied.

4.9.3.2 Relief valve. Relief valve shall show no leakage when tested to determine conformance with 3.4.6.

4.9.3.3 Manifold. Manifold shall show no sign of leakage when a back pressure of 5 lb/in^2 is applied.

4.9.4 Proof. Each flexible hose assembly with fitting, when bent to the manufacturer's recommended minimum bend radius, shall not fail or suffer permanent distortion when subjected to a hydrostatic gage pressure of 7500 lb/in^2 for a period of 2 minutes.

4.9.5 Hydrostatic. Cylinder valve assembly shall be subjected to a hydrostatic pressure of 8333 lb/in^2 using a suitable plug as a replacement for a frangible disc to determine conformance to 3.4.2. Any assembly which fails to pass the test shall be rejected.

4.9.6 Leakage rate. When the cylinder valve assembly is charged as specified in 3.6.1, the maximum leakage rate shall be not more than 3 cubic centimeters of free air per hour at 70°F ± 5°F and standard atmospheric pressure. Any assembly which fails to pass the test shall be rejected.

4.9.7 Operating force. When charged as specified in 3.6.1, the inflation assembly shall discharge when a force of not more than 20 pounds and not less than 10 pounds is applied to the actuating cable. Any assembly which fails to pass the test shall be rejected.

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4.9.8 Operational. Each valve assembly shall be tested as follows to determine conformance with 3.4.2:

(a) The cylinder valve assembly shall be connected to a manifold/hose assembly (or simulation for test purposes) incorporating the various fittings specified, after charging the cylinder as specified in 3.6.1. The entire assembly shall be chilled in a chamber until a temperature of minus 20°F ± 2°F has been reached (as indicated by attached thermocouples or other accurate temperature indicating device). When actuated, using the pull cable, the assembly shall discharge no less than 99 percent of the compressed air charge in no more than 45 seconds into the ambient minus 20°F ± 2°F chamber atmosphere. The charge shall pass through the passages of the entire inflation system assembly venting equally through the outlet diffusers of the manifold.

(b) The cylinder valve assembly shall again be tested as described in (a), except that the assembly shall be conditioned to a temperature of 165°F ± 2°F. The cylinder valve assembly shall then be discharged into the ambient 165°F atmosphere.

(c) Pull cable tension during valve actuation shall be measured to determine conformance with 3.4.2.1. Any cylinder valve assembly (component) failure shall be cause for rejection of the assembly.

4.9.9 Coupling nut torque. A torque of 180 foot-pounds shall be applied to the inflation valve at its connection to the cylinder. A torque of 20 foot-pounds shall be applied to each coupling nut of the flexible hose connecting the inflation valve to the inlet manifold. There shall be no evidence of failure, deformation, or leakage.

4.9.10 Endurance. Each cylinder valve assembly, while attached to the inflation equipment, and charged at the beginning of each cycle as specified in 3.6.1, shall be operated through five complete cylces, charge and discharge, at 70°F ± 15°F. All discharge cycles shall be tested as specified in 4.9.8, except for temperature. There shall be no evidence of malfunction or excessive wear of any part of the valve assembly. The assembled equipment shall then be subjected to the tests of 4.9.6 and 4.9.7. Any assembly which fails to pass the test shall be rejected.

4.9.11 Patch adhesion. Five of each hundred manifolds, relief valves, and topping-off valves or lesser portion thereof, shall be tested as specified in 4.9.11.1 and 4.9.11.2 to determine conformance with the requirements of 3.4.4, 3.4.5, and 3.4.6.

4.9.11.1 As received. Molded rubber patch shall not strip away from the valve or manifold. The rubber patch shall not fail below 40 pounds, when tested as follows:

(a) One-inch wide tabs, 180 degrees apart, shall be cut through the patch to the metal portion of the valve or manifold. The tabs shall be attached to a suitable test machine and pulled to failure.

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4.9.11.2 After aging. After being exposed to a temperature of 170°F ± 2°F for 72 hours ± 2 hours, the valve or manifold shall be cooled to a temperature of 75°F ± 5°F and allowed to remain at that temperature for 5 hours ± 1 hour. Tests shall be in accordance with 4.9.11.1, except that failure shall not occur below 35 pounds.

4.10 Examination of preparation for delivery. Each of the fully prepared shipping containers containing individual components of one kind and selected as a sample unit from the lot shall be visually examined in accordance with table II to determine whether packaging, packing, and marking conforms to this specification.

TABLE II. List of defects for preparation for delivery.

Item	Defects
Exterior and interior markings	Missing, incorrect, incomplete, illegible; of improper size, location, sequence, or method of application; markings not the same on the interior and exterior containers.
Packaging and packing materials	Any non-conforming component; any component missing, damaged or otherwise defective.
Workmanship	Inadequate application of the component; such as incomplete closure of the unit package, container flaps, loose strapping, bulging or distortion of the containers.
Exterior and interior weight or content	Gross or net weight exceeds the requirement; more than one kind in the same container.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of the referenced documents listed in section 2, see 6.6.)

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2.1).

5.1.1 Level A. Each inflation unit and associated parts thereto shall be individually unit protected and shall be preserved-packaged in accordance with method III of MIL-P-116. Openings in the unit's valve, manifold ends, hose or fittings shall be sealed with plugs or caps. Each inflation unit and associated parts unit protected as specified herein shall be consolidated and packaged in a snug fitting fiberboard box conforming to PPP-B-636, class domestic, minimum 275 pound Mullen test. Each unit shall be cushioned, blocked or braced within the container to insure integrity of the inflation valve mechanism. Loose fill material such as polystyrene forms, wadded or shredded paper, sawdust and similar materials shall not be used as the cushioning or blocking media. Containers shall be closed in accordance with method I as specified in the appendix to the box specification.

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5.1.2 Level C. Preservation-packaging of assembled inflation units shall afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity for immediate use. The contractor's normal retail or wholesale preservation-packaging methods may be utilized when they meet the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2.1).

5.2.1 Level A. Inflation units preservation-packed as specified in 5.1 shall be packed in wood, wood cleated, or fiberboard boxes conforming to PPP-B-621, class 2, PPP-B-601 overseas type PPP-B-640, class 2, or PPP-B-636, class weather resistant, respectively. Container selection shall be at the contractor's option. Wood and wood cleated boxes shall have case liners conforming to MIL-L-10547. Case liners shall be closed and sealed in accordance with MIL-L-10547. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto, with method V applicable to PPP-B-636 containers. PPP-B-636 and PPP-B-640 fiberboard containers shall be reinforced with pressure-sensitive reinforced tape or non-metallic strapping in lieu of steel straps in accordance with the applicable specification.

5.2.2 Level B. Inflation units preservation-packaged as specified in 5.1 shall be packed in containers as specified in 5.2.1, except that the containers shall be of the domestic type or class, and case liners are not required. Containers shall be closed in accordance with the applicable container specification or appendix thereto, with method I closure applicable to PPP-B-636 containers.

5.2.3 Level C. Inflation units preservation-packaged as specified in 5.1 shall be packed in containers acceptable to the common carrier which will insure safe delivery to destination in a satisfactory condition at the lowest applicable rate. Containers, packing, and method of shipment shall comply with Uniform Freight or National Motor Freight Classification Rules and Regulations or other carrier rules as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order (see 6.2.1), interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129 and shall include the date of manufacture (month and year).

5.3.1 Special marking. Filled cylinders, interior containers housing same, and the exterior shipping container shall contain a green label in accordance with the Code of Federal Regulations, Title 49, CFR 170-199. In addition, the above shall be marked "NON-FLAMMABLE COMPRESSED GAS" and "COMPRESSED AIR". The size and color of the markings shall conform to DOT Rules and Regulations as specified herein. The markings shall be conspicuous and legible.

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6. NOTES

6.1 Intended use. The inflation equipment covered by this specification is intended for use as the inflation system for lifeboats.

6.1.1 Type I. With cylinder capacity of 250 in^3, this equipment is used for 15-person MK 5 lifeboats.

6.1.2 Type II. With cylinder capacity of 455 in^3, this equipment is used for 25-person MK 6 lifeboats.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Assembly type including cylinder capacity (cylinder only) (see 1.2 and 3.4.1).
(c) Quantities of special recharging fittings and tools, if required (see 3.4.2.6).
(d) If pressure tests are to be conducted under conditions other than specified (see 4.8.2).
(e) Levels of preservation-packaging, and packing required (see 5.1 and 5.2).
(f) Special marking required (see 5.3).

6.2.2 Data requirements. When this specification is used in a contract which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulation (DAR), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID	Option
4.5.1.1	First article inspection report	DI-T-4902	---

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract, regardless of whether an identical item has been supplied previously (for example, test reports).

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6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List QPL-24490 whether or not such products have actually been so listed by that date. The attention of the contractors is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.4 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.5.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.5.2 It is recommended that the preservation-packaging and packing of repair parts be in accordance with MIL-R-196 for the level as specified

Destination	Level Preservation-packaging	Packing
On board	A	C
Stock	A	B
Immediate use	C	C

Repair parts should be packaged one part per unit package, except that all parts comprising a single set or assembly should be individually protected and packaged together. When unit packaged as a set, assembly, or quantities greater than one, each item should be wrapped or cushioned to prevent direct surface contact with surface of adjacent parts.

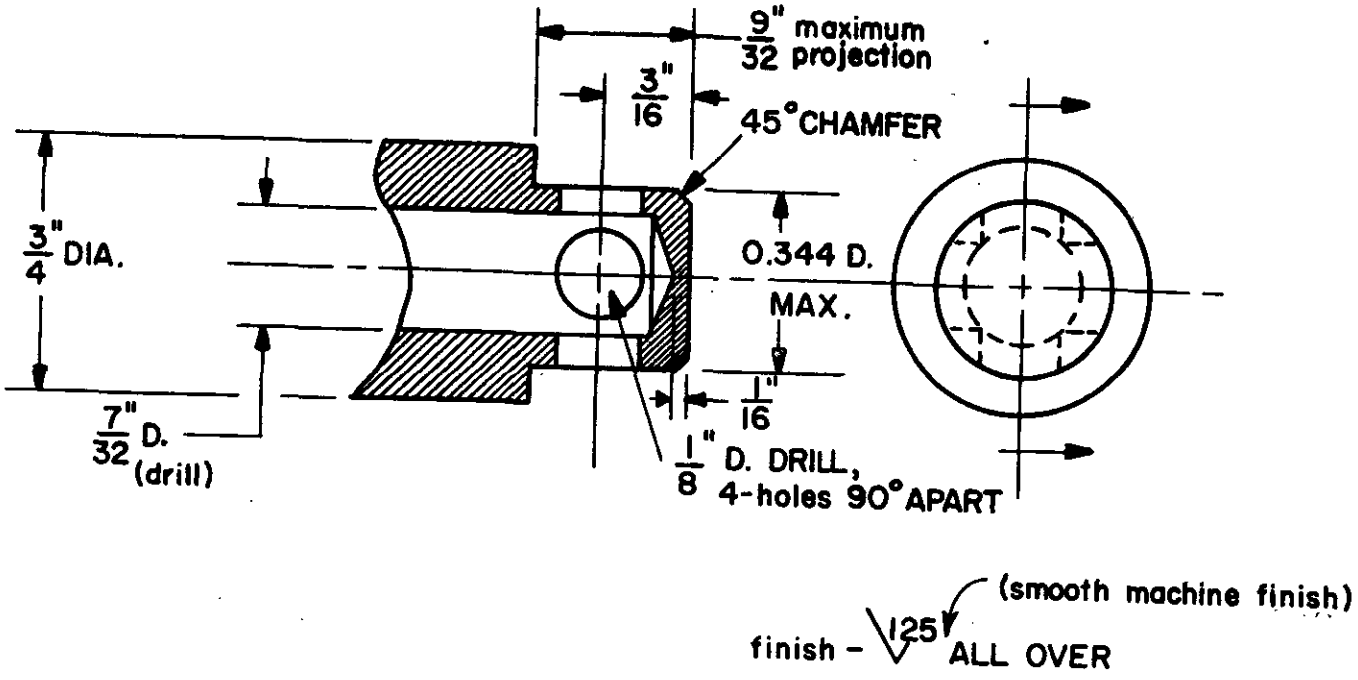
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6.6 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 Changes from previous issue. Asterisks (*) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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FIGURE 1. Anti-recoil fittings.

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